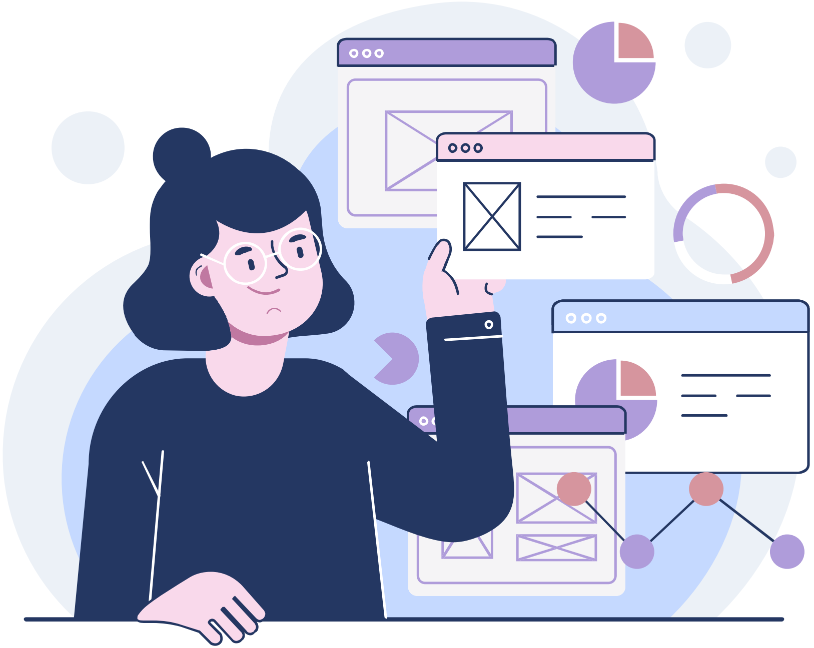
How can Rossman optimise store operations to improve sales?

By Angela, Felix, Jasmine, Nicole, Yanghan



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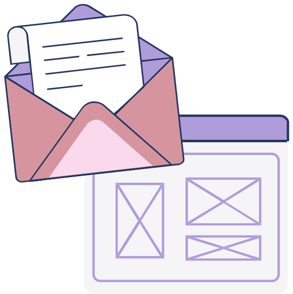
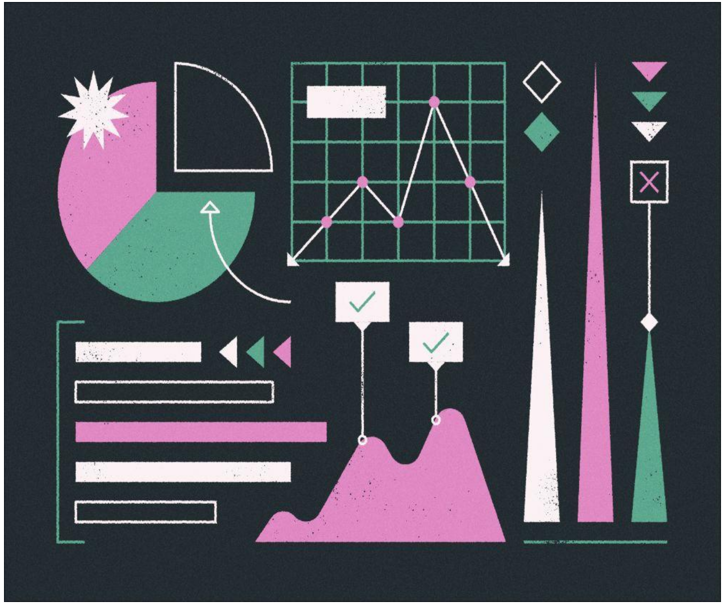
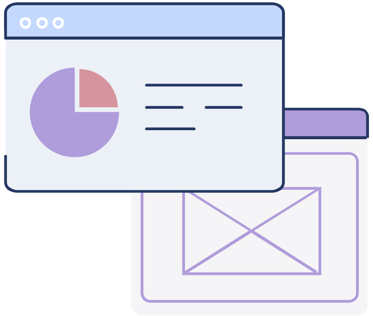




01 Introduction



Background

- Rossmann: 3,000+ drugstores in 7 European countries
- Sales depend on promotions, competition, holidays, seasonality
- Accurate forecasting → better staffing, inventory, budgeting
- Project goal: clean, analyze, and model sales data for insights

Data Overview

Store	DayOfWeek	Date	Sales	Customers	Open
1	5	2015-07-31	5263	555	1
2	5	2015-07-31	6064	625	1
3	5	2015-07-31	8314	821	1
4	5	2015-07-31	13995	1498	1
5	5	2015-07-31	4822	559	1
6	5	2015-07-31	5651	589	1
7	5	2015-07-31	15344	1414	1
8	5	2015-07-31	8492	833	1

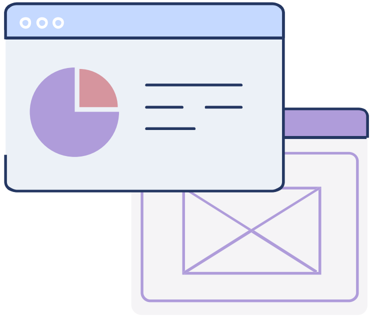
train.csv

- daily sales for 1,115 stores (2013–2015)

store.csv

- store details (type, assortment, competition, promos)

StoreType	Assortment	CompetitionDistance	CompetitionOpenSir	CompetitionOpenSir
c	a	1270	9	2008
a	a	570	11	2007
a	a	14130	12	2006
c	c	620	9	2009
a	a	29910	4	2015
a	a	310	12	2013
a	c	24000	4	2013
a	a	7520	10	2014



02

Data Wrangling

Dropped closed store days

A large number of rows had zero sales, which would have biased the model towards predicting zeros.

Merged train with store data

Merged datasets based on store IDs to add features such as type, assortment, promos and competition.

Handling missing values

Competition distance filled with median to keep realistic distribution. Promo2field set to 0 as store was not participating. Competition open dates filled with 0, treated as no competition information.

Converted dates

Captures sales patterns from Monday to Friday and allows for time series analysis

Random Sampling

Sampled 20% of stores randomly rather than picking the top 20% to avoid overweighting big stores

Weather Data

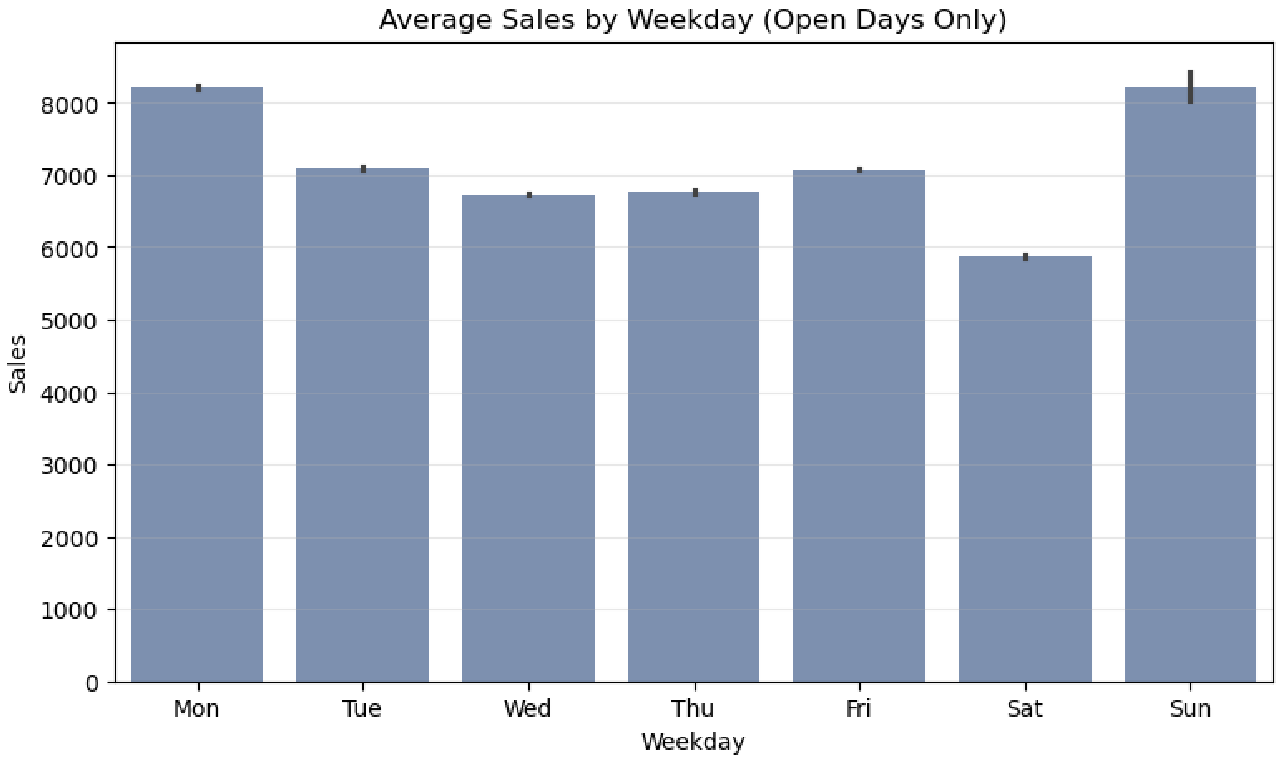
Imported and merged weather data from Kaggle because it was an external factor affecting sales.

03

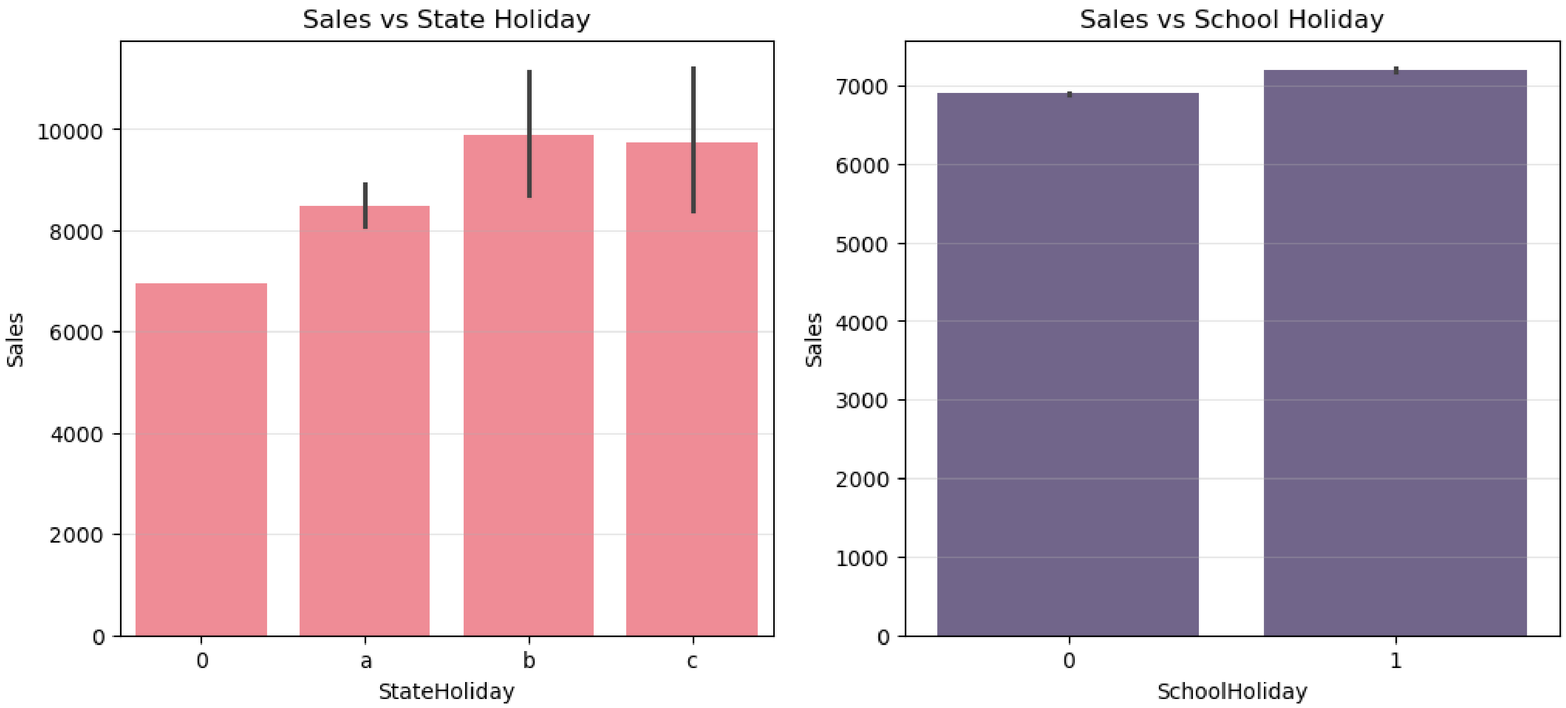
Exploratory Data Analysis

How do sales vary across days of the week and how are they affected by holidays?

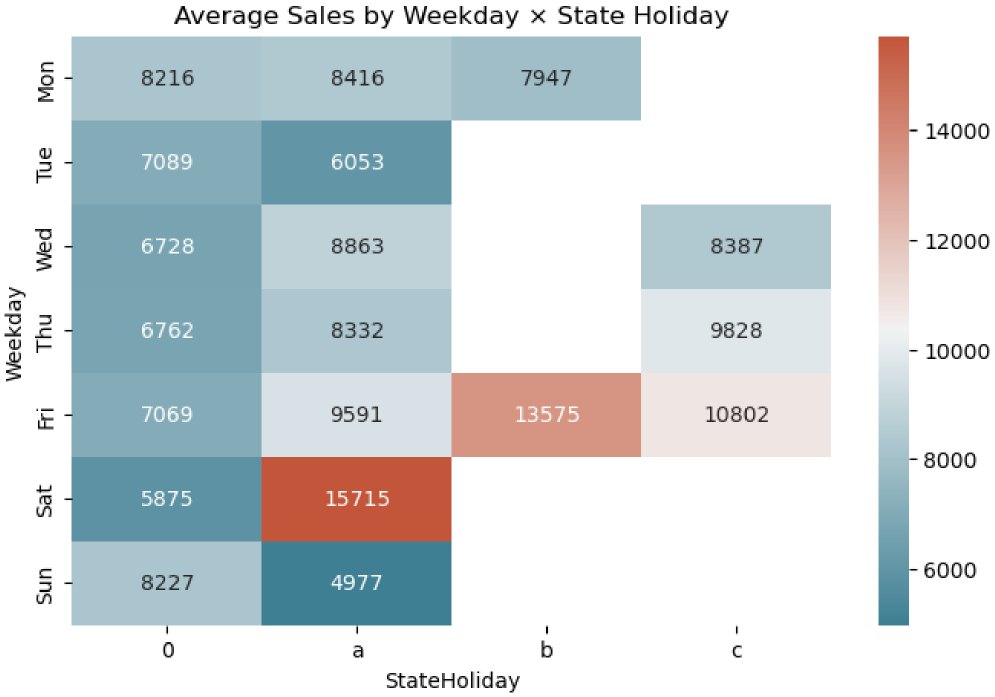
Average Sales by Weekday



Average Sales by Holiday



Average Sales on Weekdays x State Holiday



Key Insights for Rossman:

1. Increase staff and stock replenishment on **Mondays, Fridays** and **weekends with holidays** have highest demand

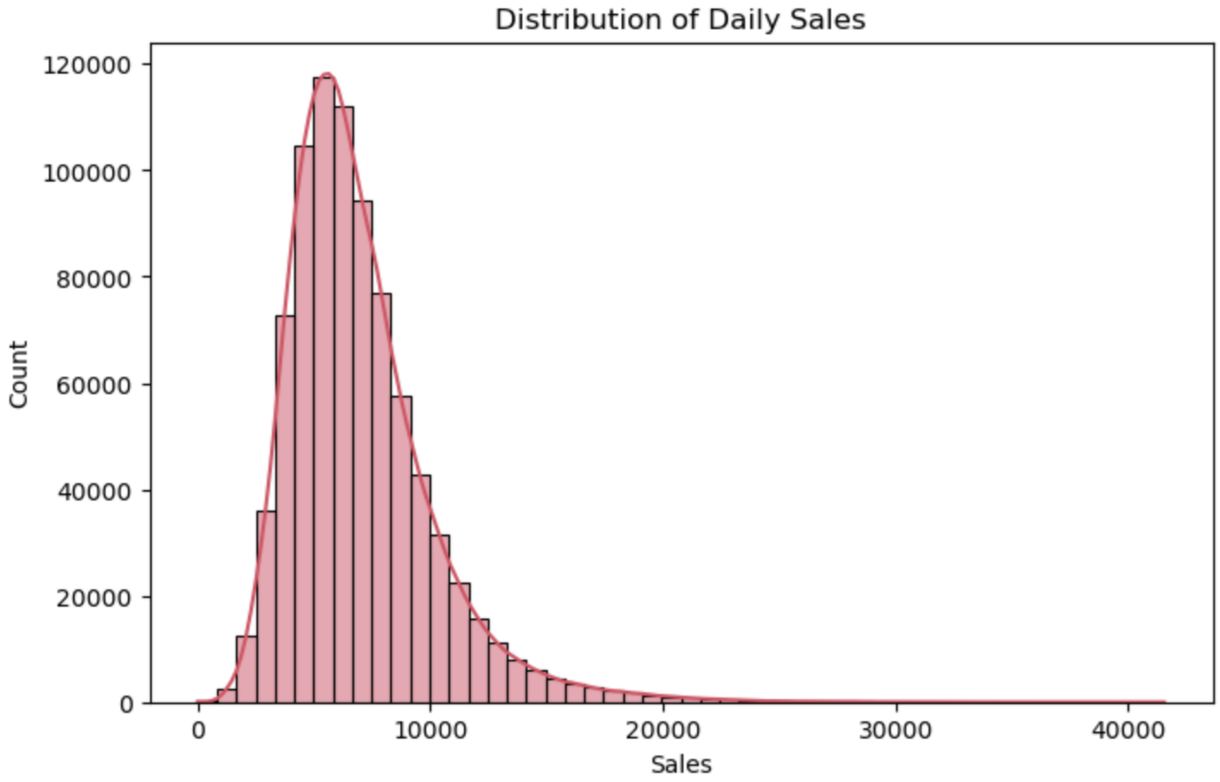
2. Promotion campaigns on **Tuesdays to Thursday, and Saturdays** to promote sales

3. Keep only high-performing **Sunday** stores open. Increase staff and stock replenishment for open stores.

When?

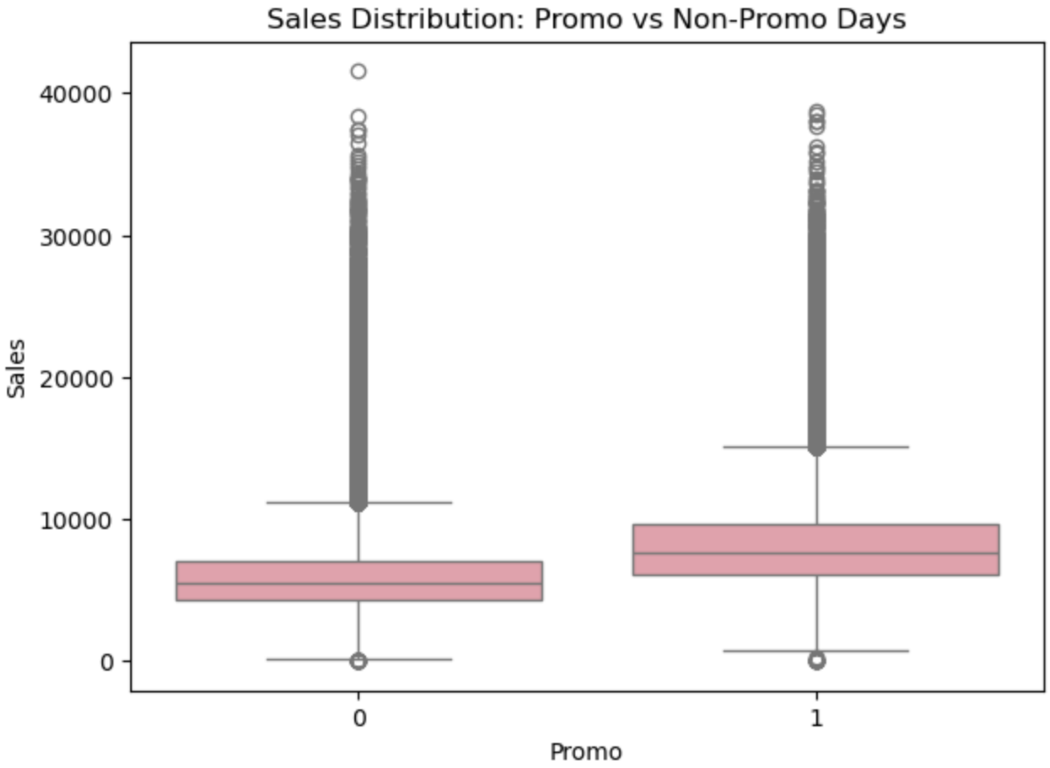
is the best time to run promotions

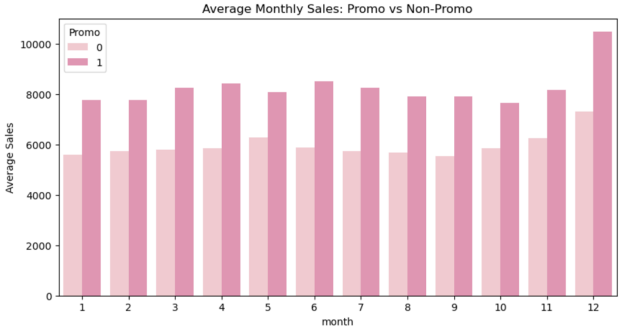
Distribution of Daily Sales



- Most sales between 5k–10k, occasional spikes
- Promotions shift sales into the higher tail

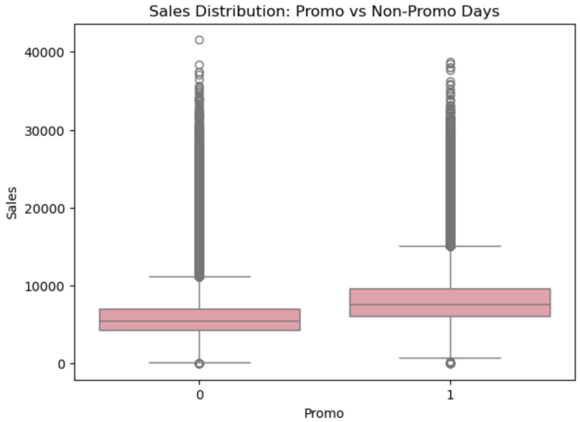
Promo Effectiveness Across the Year

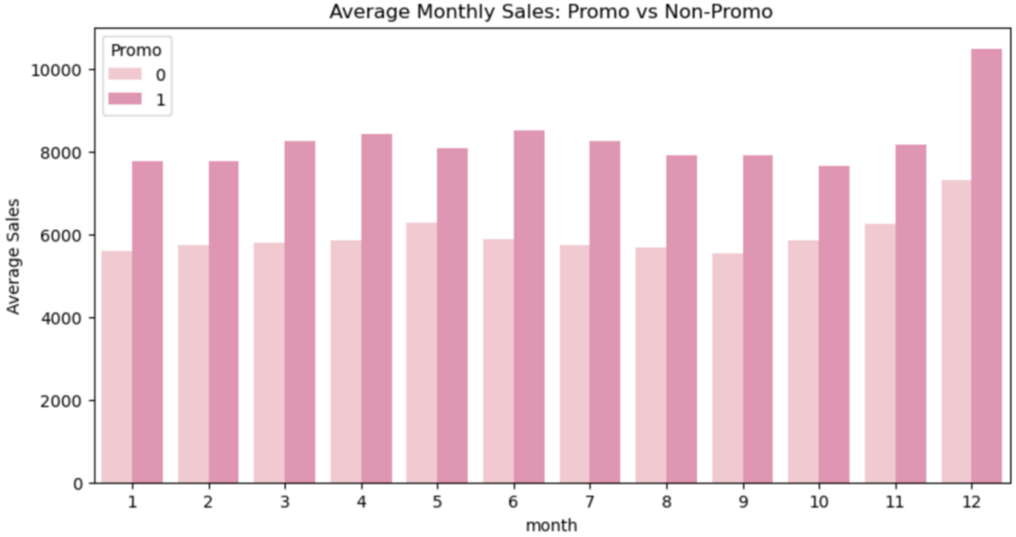




- Promotions reliably increase sales
- December = strongest promo season
- Mid-year promos still provide lift

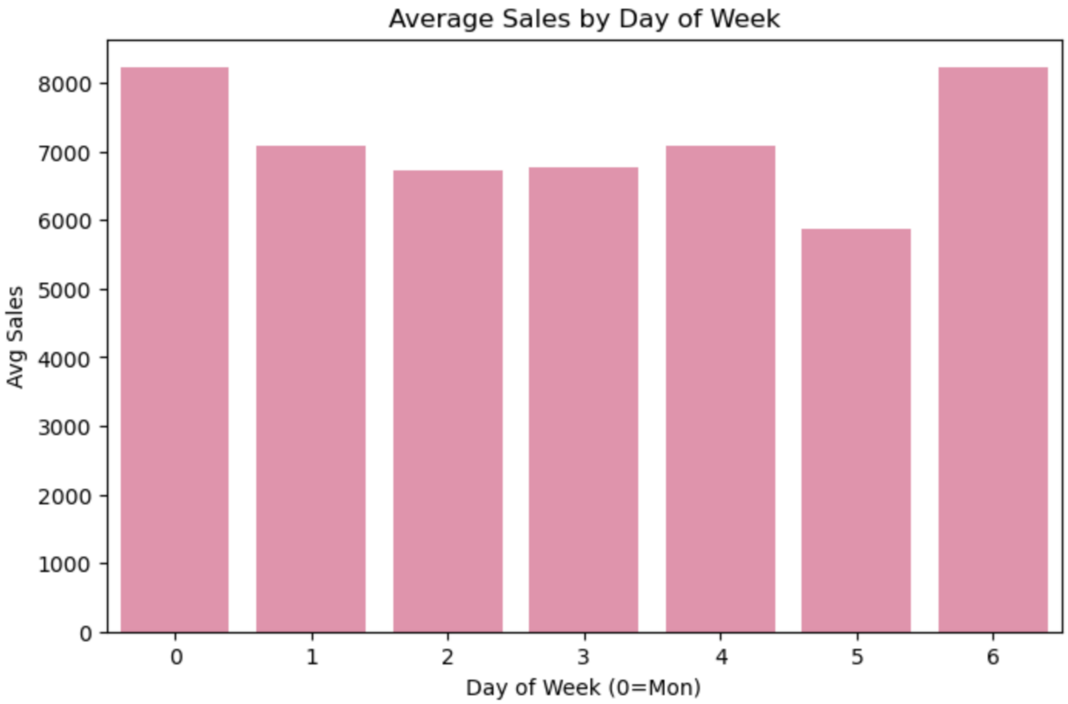
Promo Effectiveness Across the Year





- Promotions reliably increase sales
- December = strongest promo season
- Mid-year promos still provide lift

Average Sales by Day of Week

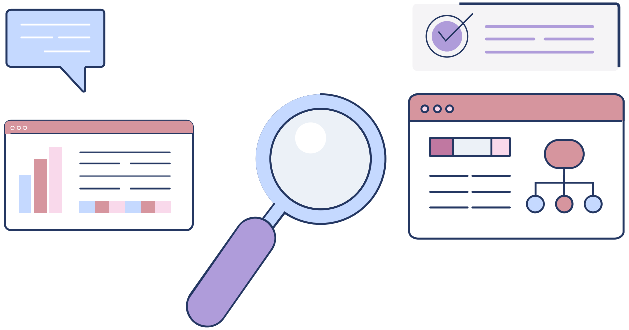


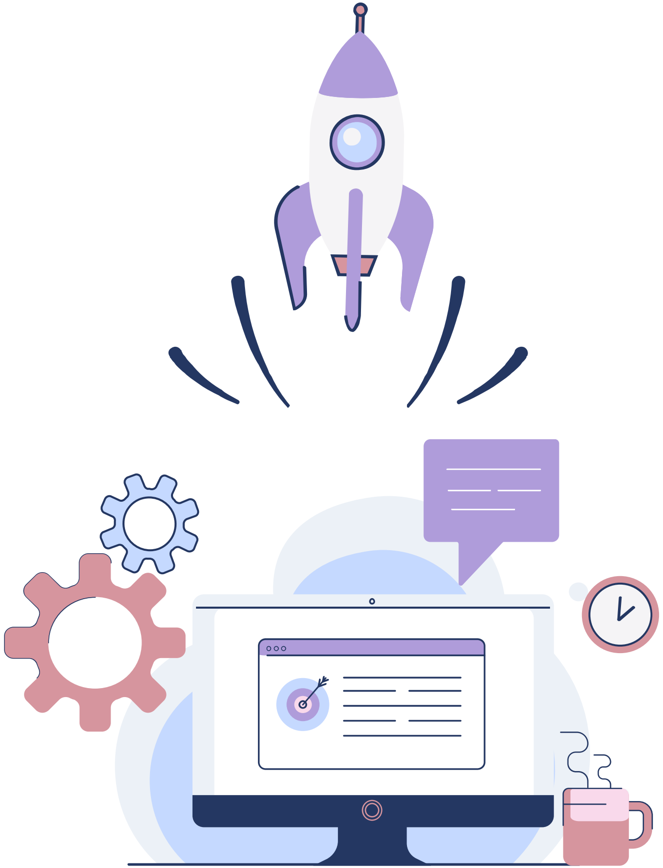
- Sales peak on Sundays & Mondays
- Saturday is weakest day

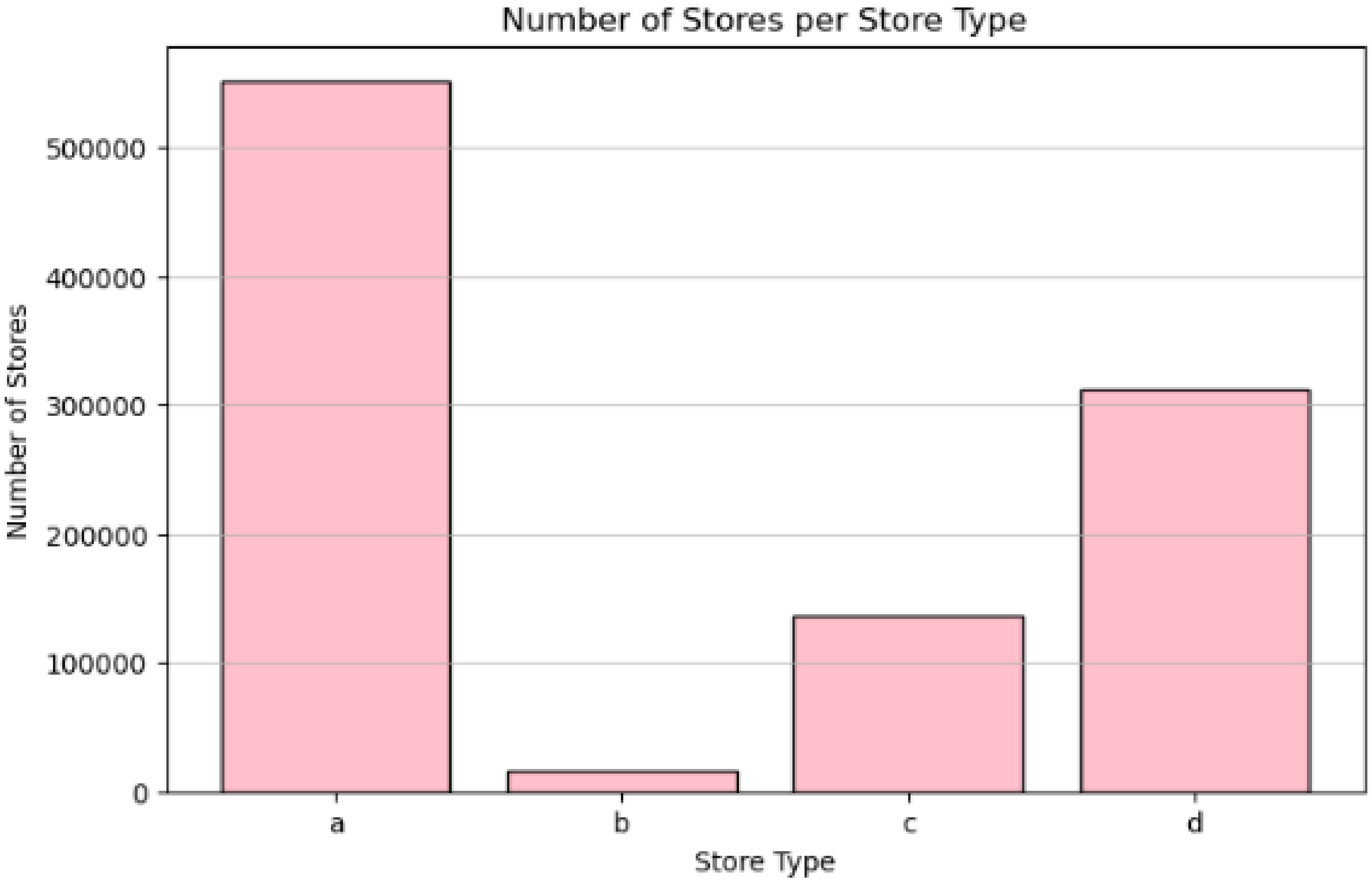
Key Insights for Rossman:

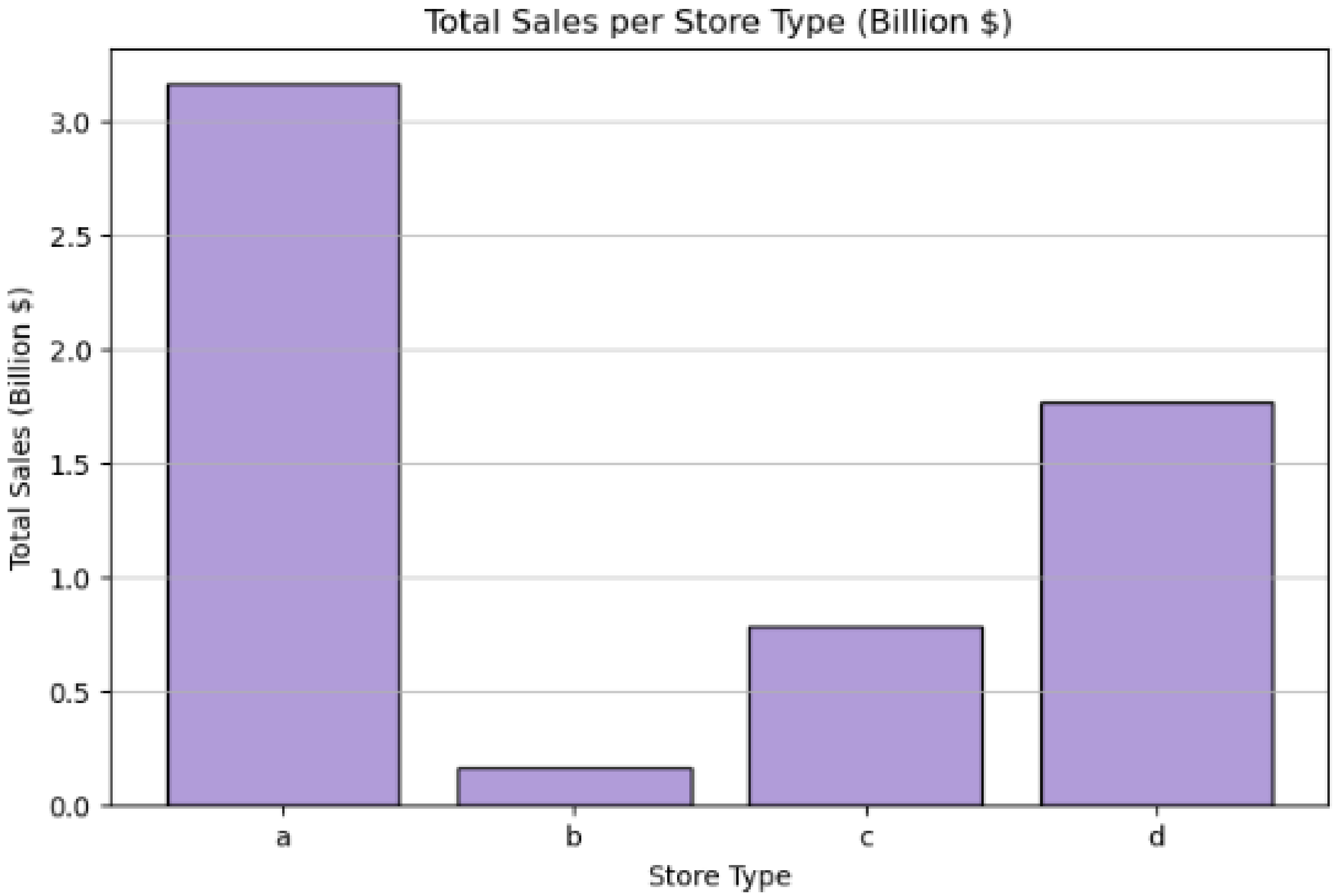
1. Promotions boost sales on average
2. December is the most effective month
3. Mid-year promos help sustain sales
4. Sundays & Mondays are best promo days

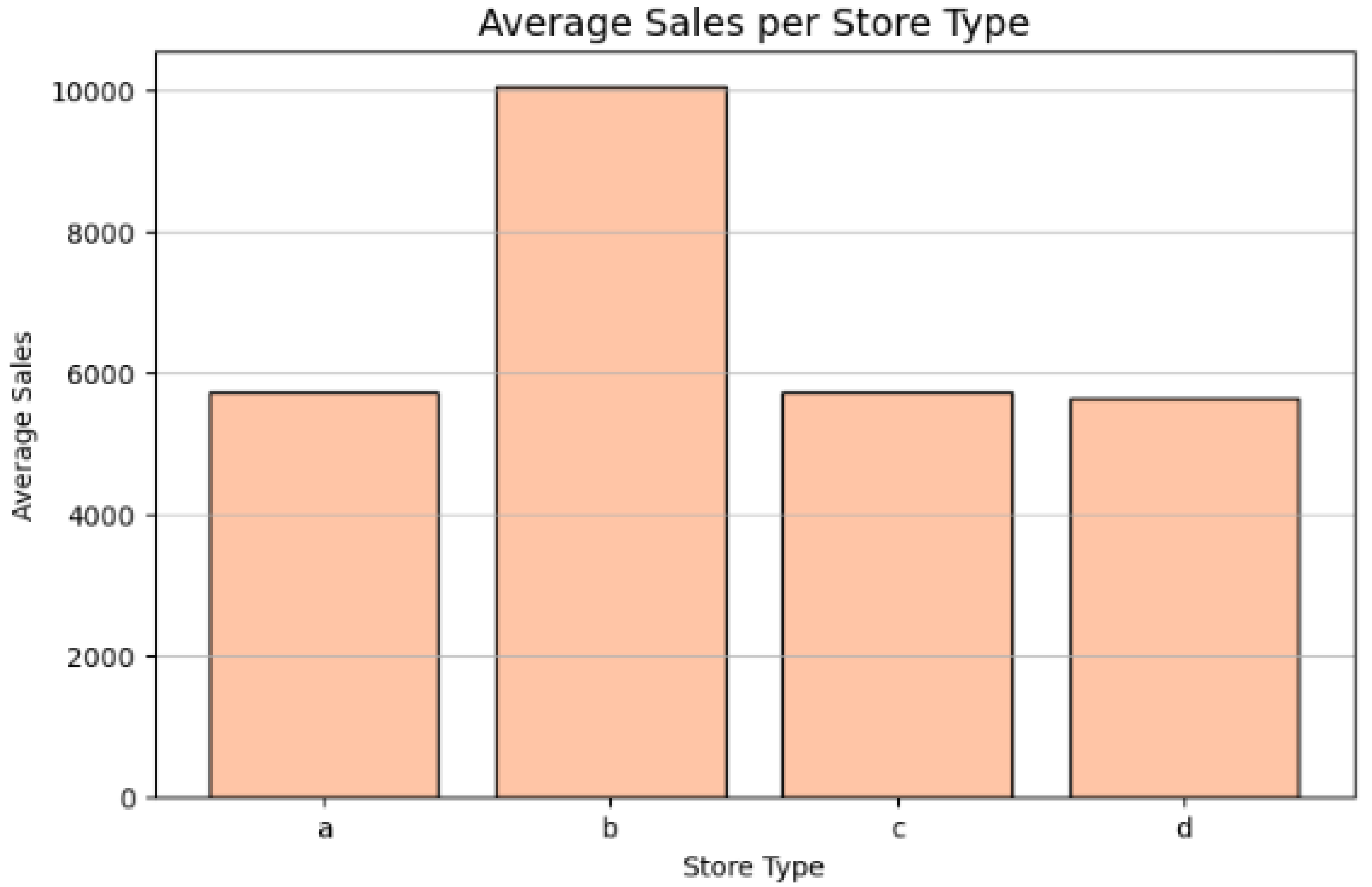
Does store type significantly explain the differences in sales performance among stores?



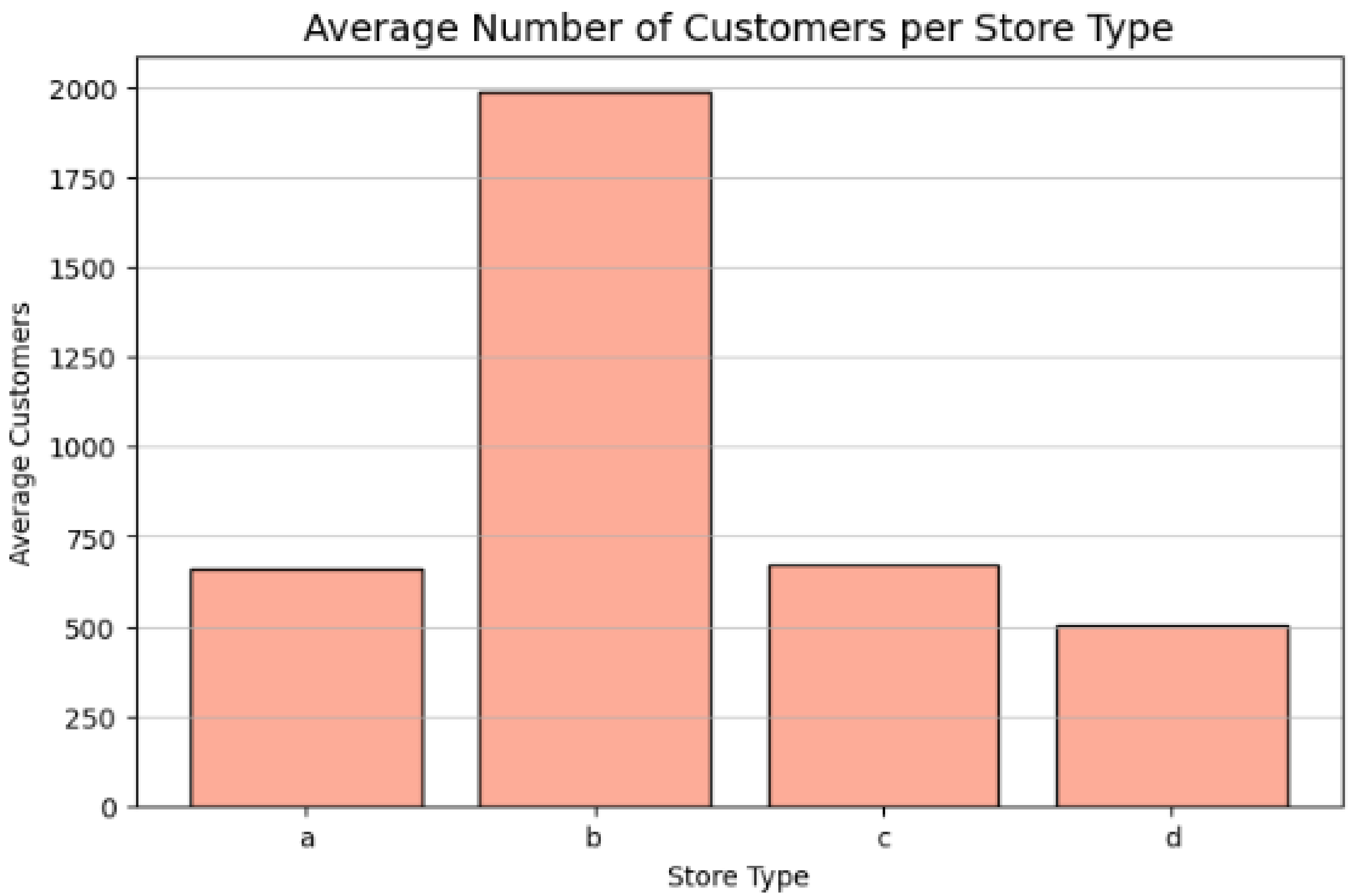


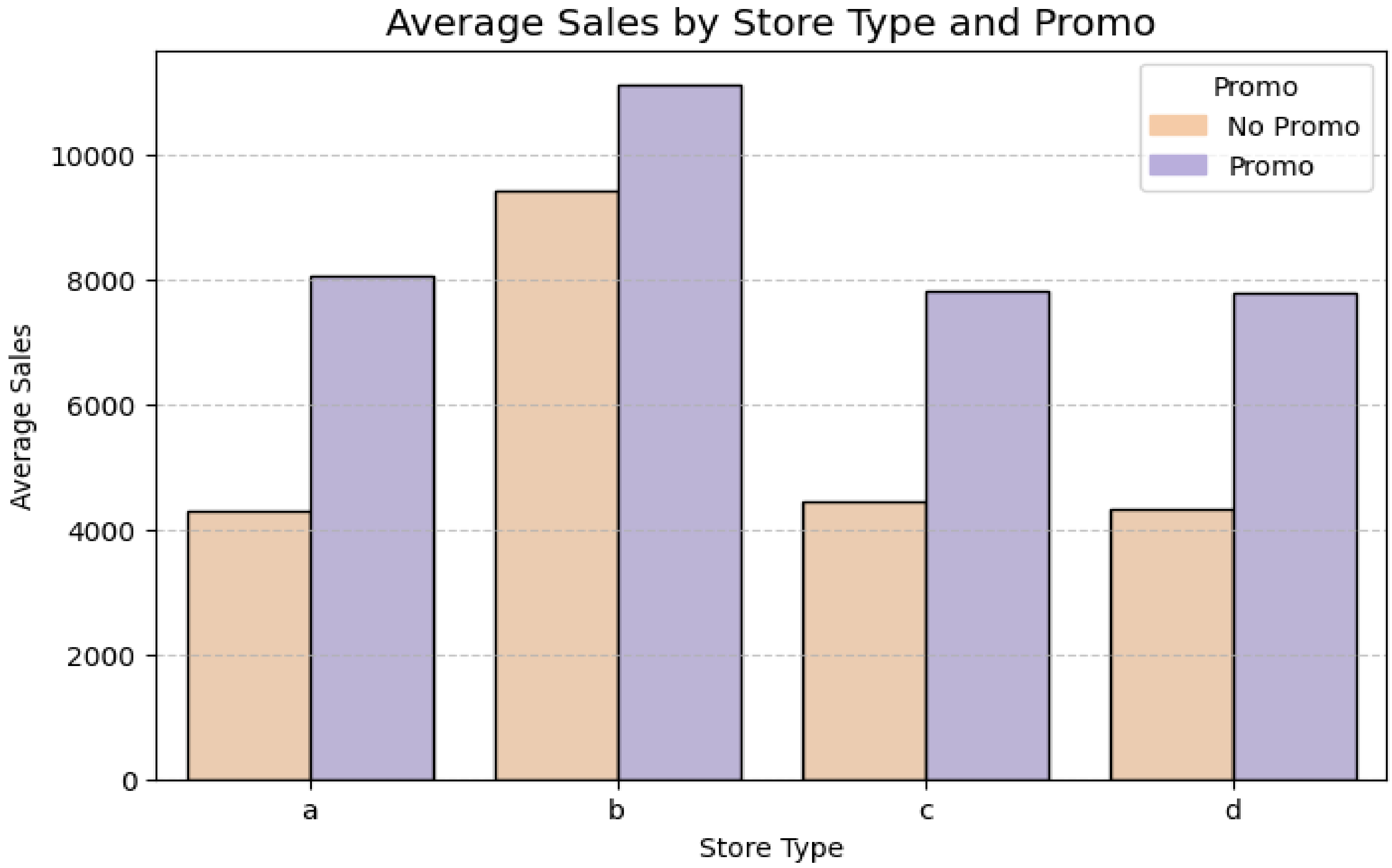

Number of Stores per Store Type
Number of Stores
0
100000
200000
300000
400000
500000
a
b
c
d
Store Type


Total Sales per Store Type (Billion $)
Total Sales (Billion $)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
a
b
c
d
Store Type



Average Sales per Store Type
Average Sales
10000
8000
6000
4000
2000
0
a
b
c
d
Store Type


Average Number of Customers per Store Type
Average Customers
2000
1750
1500
1250
1000
750
500
250
0
a
b
c
d
Store Type


Average Sales by Store Type and Promo
Promo
No Promo
Promo
10000
8000
6000
4000
2000
0
Average Sales
a
b
c
d
Store Type

Key insights for Rossman

1. Study and replicate Type B best practices

1. Boost customer traffic in Types A, C, D through promotions/loyalty

1. Allocate resources by store type ROI

1. Pilot store conversions to Type B model

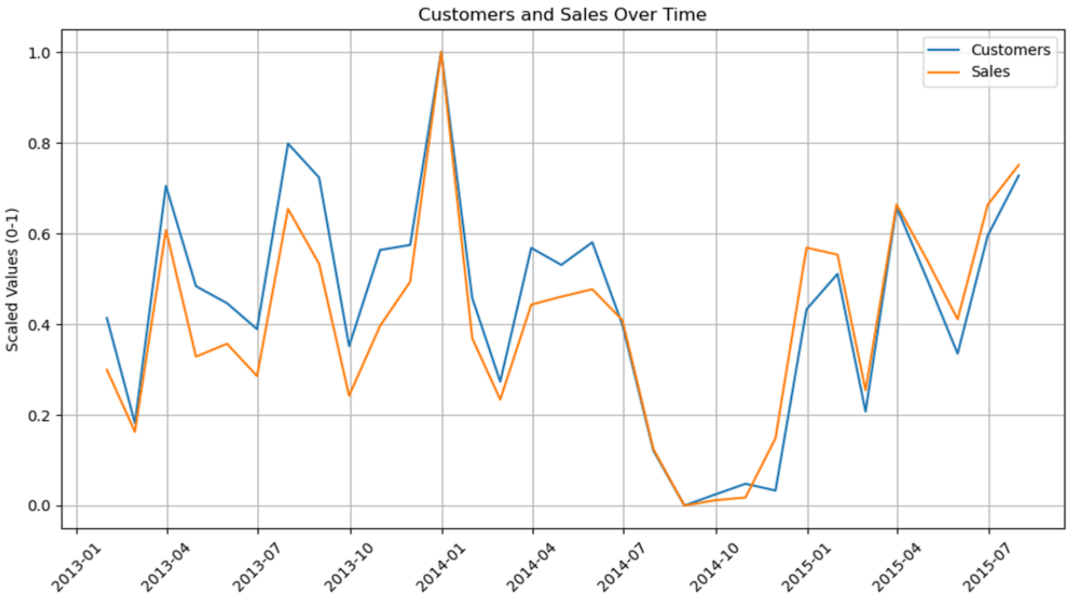
04

Predictive Modelling: Can external factors like weather affect sales?

Baseline model on customers only

Why customers?

A quick heatmap demonstrates that customers has by far the greatest effect on sales, so as a baseline model, we used linear regression and found the model to have a RMSPE of 29%, which would place us 3000th in the kaggle competition. The goal now is to improve this score.

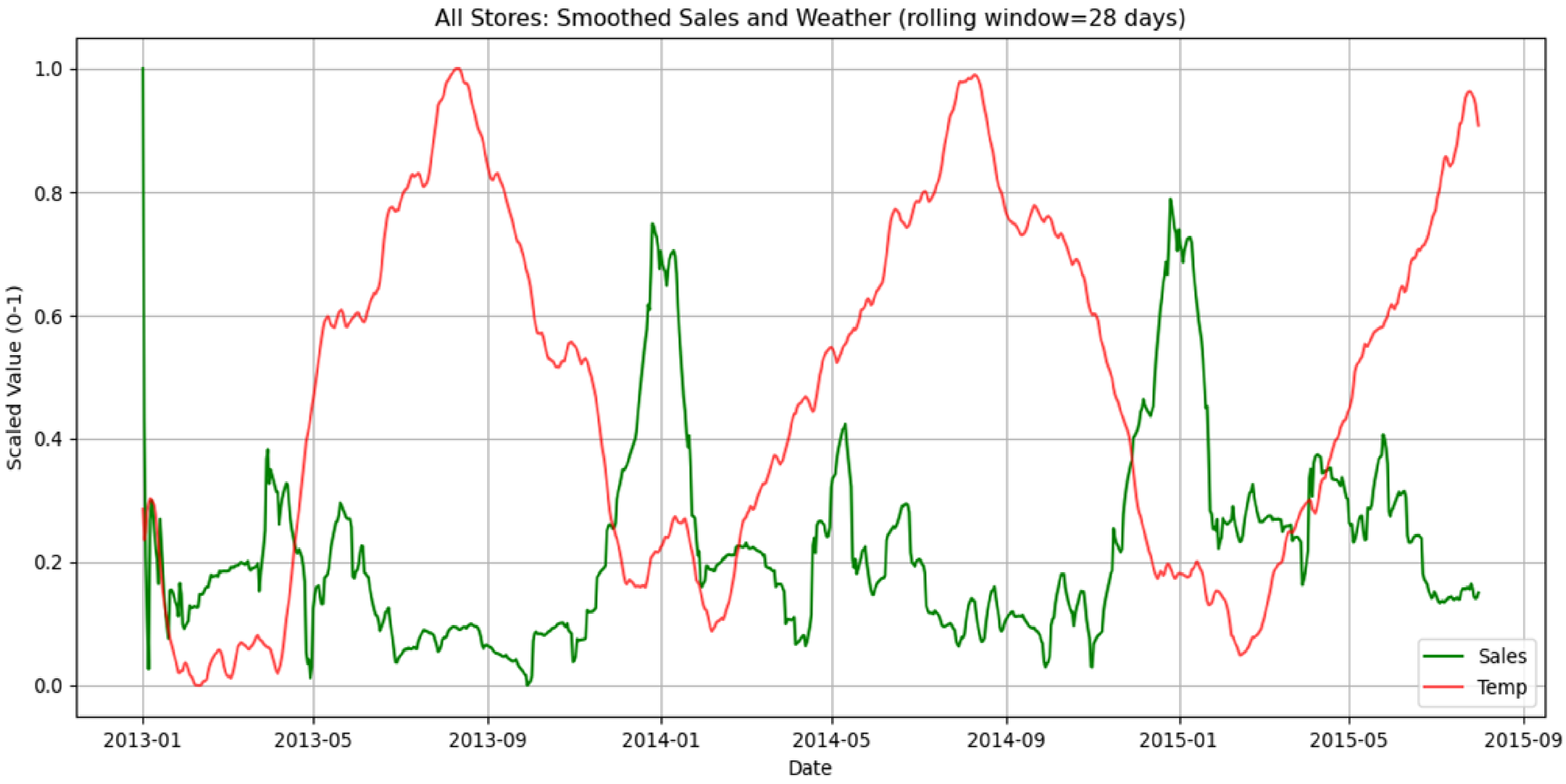


Changes to original model

1. A random 20% of the data was chosen (as top 20% for example would only represent the stores in the most densely populated areas).

1. Storetype and assortment were OneHotEncoded, so they could be included in the model

1. Weather data from the Kaggle competition was included to see the effect of weather on the sales.


All Stores: Smoothed Sales and Weather (rolling window=28 days)
Scaled Value (0-1)
1.0
0.8
0.6
0.4
0.2
0.0
2013-01
2013-05
2013-09
2014-01
2014-05
2014-09
2015-01
2015-05
2015-09
Date
Sales
Temp

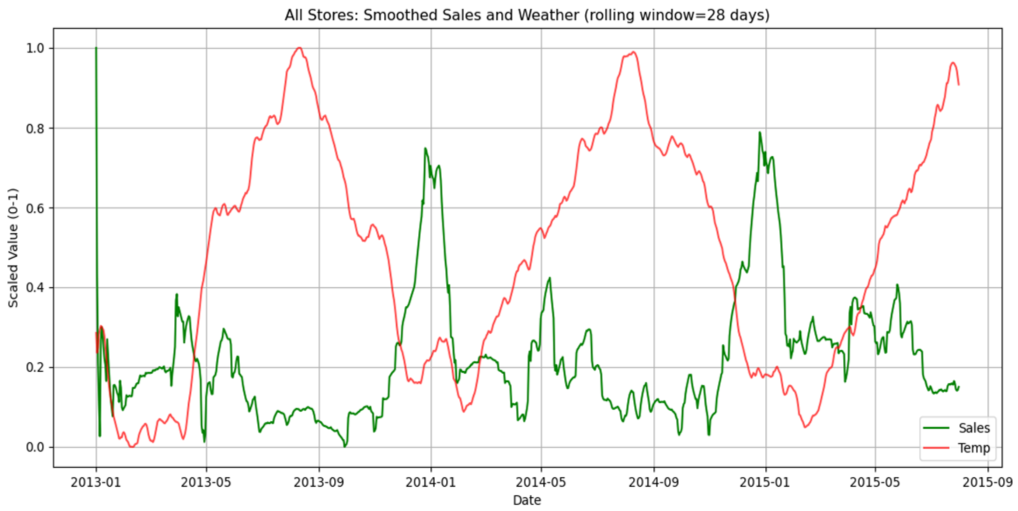
Other ways to increase sales

(Possibility unethical)

- Raise prices of in demand items in the winter months e.g cold and flu medication.

- The more in demand the more you can spike the price.

- Likewise flexibly changing the prices of drinks such as electrolytes based on the forecast (e.g hot couple days)



Results Table - RMSPE

	Degree 1	Degree 2	Degree 3	Degree 4
Baseline Model (customers)	29	23	22	23
Addition of weather and competition	28	21	19	20
Addition of storetype and assortment	19	17	16	15

In future

- I would like to use cross validation and lasso regularisation to see the variability of the results and see which variables to potentially remove.

- Right now for example I have multiple weather variables such as humidity and temperature, by determining the impact of each variable I could see if the weather variables are impacting the model or just each other.

- Thus, the store can see which features to target to increase sales

Thanks from group 5!

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